



Incident report analysis

Summary	The multimedia company experienced a Denial of Service (DoS) attack that disrupted internal network operations for approximately two hours. The attack was caused by a flood of ICMP packets sent by a malicious actor through an unconfigured firewall. The excessive traffic overwhelmed network resources, preventing legitimate users from accessing network services. The incident response team mitigated the attack by blocking incoming ICMP packets, taking non-critical services offline, and restoring critical services. Following the investigation, several security controls were implemented to reduce the risk of similar attacks in the future.
Identify	The cybersecurity team conducted an assessment of the affected network infrastructure and identified an improperly configured firewall as the primary vulnerability that allowed the attack to occur. Key findings included: • The firewall lacked rules to limit or filter ICMP traffic. • The internal network infrastructure and critical services were vulnerable to traffic flooding attacks. • Network availability was affected, preventing employees from accessing business resources. • Existing monitoring capabilities were insufficient to detect abnormal ICMP traffic before service disruption occurred. • Critical assets affected included network infrastructure, servers, and internal communication systems.

Protect	To strengthen security and reduce the likelihood of future DoS attacks, the organization implemented several protective measures: • Configured firewall rules to limit the rate of incoming ICMP packets. • Enabled source IP address verification to identify and block spoofed IP addresses. • Updated firewall configurations and security policies to ensure proper network protection. • Established regular firewall audits and configuration reviews. • Provided cybersecurity awareness training for IT personnel regarding secure network configuration and DoS mitigation techniques. • Implemented network segmentation to help isolate critical systems from potential attacks. • Developed procedures for regularly reviewing and testing security controls.
Detect	The organization enhanced its ability to identify suspicious activity and potential attacks through improved monitoring and detection mechanisms. Detection improvements include: • Deployment of network monitoring software to identify unusual traffic patterns. • Implementation of an Intrusion Detection System (IDS) and Intrusion Prevention System (IPS) to monitor and filter suspicious ICMP traffic. • Configuration of automated alerts for excessive ICMP traffic and potential DoS indicators. • Continuous monitoring of firewall logs and network activity.

	• Development of baseline network traffic patterns to quickly detect anomalies.
Respond	The incident response team successfully contained and mitigated the attack through coordinated actions. Response activities included: • Blocking incoming ICMP traffic responsible for the attack. • Taking non-critical services offline to preserve resources for critical operations. • Restoring essential network services as quickly as possible. • Investigating the source and method of the attack. • Communicating incident status and recovery efforts to management and affected personnel. • Documenting lessons learned and updating incident response procedures. • Conducting a post-incident review to identify opportunities for improvement.
Recover	After the attack was contained, the organization focused on restoring normal operations and strengthening future resilience. Recovery actions included: • Restoring all critical network services and returning systems to normal operation. • Verifying the integrity and functionality of network infrastructure following recovery.

	• Monitoring systems closely to ensure no recurring attack activity occurred. • Reviewing and validating newly implemented firewall rules and security controls. • Updating disaster recovery and business continuity plans to address future DoS incidents. • Scheduling regular security assessments and testing of recovery procedures. • Communicating recovery completion and system status updates to employees and management.
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Reflections/Notes:

This incident demonstrated the importance of properly configured security controls and proactive network monitoring. A single firewall misconfiguration allowed a malicious actor to disrupt business operations significantly. By implementing stronger firewall policies, traffic filtering, IDS/IPS technologies, and continuous monitoring, the organization has improved its security posture. Moving forward, regular security audits, employee training, and continuous review of network defenses will be essential to minimizing the risk of future DoS attacks and ensuring business continuity.